Quantification: 535_z20c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	14
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3123.64	2390.5	2354.01	px^2
major_axis_length_px	lacuna shape	91.5498	84.2904	37.8068	px
minor_axis_length_px	lacuna shape	45.0262	40.768	18.6509	px
aspect_ratio	lacuna shape	2.1221	2.0343	0.639	unitless
eccentricity	lacuna shape	0.8338	0.8709	0.1326	unitless
solidity	lacuna shape	0.8092	0.8206	0.1146	unitless
roots_count	headline	8.9286	7.5	4.1596	count
ring_length_r30_px	headline	301.071	294	89.9354	px
ring_length_r60_px	reported	646.5	671.5	188.478	px
owned_length_px	ownership-dependent	984.519	864.76	583.225	px
edge_count	ownership-dependent	42.3571	35.5	25.4214	count
mean_edge_length_px	ownership-dependent	24.2673	21.9537	7.3813	px
perimeter_px	normalised	297.172	252.664	181.059	px
ring_area_r30_px2	normalised	10045.9	9306	3454.95	px^2
ring_area_r60_px2	normalised	23176.9	22719.5	6825.42	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9997	1	0.0012	unitless
ring_density_r30	normalised	0.0307275	0.029295	0.00623879	px^-1
ring_density_r60	normalised	0.0283421	0.0294762	0.00624844	px^-1
roots_per_100px_perimeter	normalised	3.2105	3.2382	0.9265	per 100 px
ring_attached_length_r30_px	network v2	266.071	254.5	97.4454	px
ring_attached_length_r60_px	network v2	484.429	478	171.672	px
ring_length_w_r30_px	network v2	341.194	332.271	101.018	px
ring_length_w_r60_px	network v2	733.946	756.597	215.03	px
sholl_crossings_r10	network v2	7.0714	6	2.9733	count
sholl_crossings_r20	network v2	9.4286	9	3.1796	count
sholl_crossings_r30	network v2	8.5714	8.5	2.6808	count
loop_count	loops, report only	1.7857	0.5	2.517	count
loop_edge_share	loops, report only	0.1071	0.0625	0.1224	share
ring_width_mean_r30_px	width	7.8413	7.7833	0.898	px

Per field
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measure	value	units
analysed_area_px2	1.00484e+06	px^2
total_skeleton_length_px	22825	px
canalicular_length_density_per_px	0.022715	px^-1
junction_count	363	count
junction_density_per_px2	0.00036125	px^-2
lacuna_count	14	count

lacunae_per_px2	1.393e-05	px^-2
skeleton_component_count	398	count
mean_component_length_px	57.3492	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0255804	px^-1
field_density_without_flagged_per_px	0.0222829	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	586	px
unattached_component_count	36	count
canalicular_length_density_attached_per_px	0.022715	px^-1
field_length_density_with_unattached_per_px	0.0232981	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	26	count
loop_edge_share	0.0884	share
edge_count	1108	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.59	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	11.3137	px
width_pixels_used	19780	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3380	83.9727	56.9872	1.4735	0.7345	0.8272	-0.9852	92.8941	716.001	False
2	532	31.4193	21.8844	1.4357	0.7175	0.9449	-0.5795	222.887	852.859	False
3	1982	85.686	32.1789	2.6628	0.9268	0.8251	-1.1473	243.598	120.229	False
4	1956	84.6082	30.6094	2.7641	0.9323	0.8815	-0.4607	416.865	210.624	False
5	7294	147.859	76.1316	1.9421	0.8573	0.7458	-0.0651	474.999	284.411	False
6	4856	115.922	57.0373	2.0324	0.8706	0.8161	-1.3108	461.194	574.536	False
7	3706	78.2414	69.9705	1.1182	0.4475	0.789	0.5114	581.897	502.58	False
8	2088	68.5717	40.0641	1.7115	0.8116	0.9198	0.595	570.565	144.556	False
9	2662	104.173	41.4719	2.5119	0.9173	0.6642	1.1534	613.184	873.72	False
10	2280	82.4791	35.9781	2.2925	0.8998	0.9205	-1.1889	632.433	317.15	False
11	1049	49.7107	28.8293	1.7243	0.8147	0.9468	1.1742	770.511	315.037	False
12	8599	182.211	74.2326	2.4546	0.9133	0.6858	-1.2762	833.044	641.531	False
13	2501	85.9153	42.1925	2.0363	0.8711	0.8016	1.0022	846.978	318.042	False
14	846	80.9285	22.7986	3.5497	0.9595	0.5599	-0.1838	864.301	863.892	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	351	834	857.139	29	29.5565	7.4729
2	5	203	544	585.428	24	24.3929	7.5355
3	9	398	928	2255.04	104	21.6831	8.2992

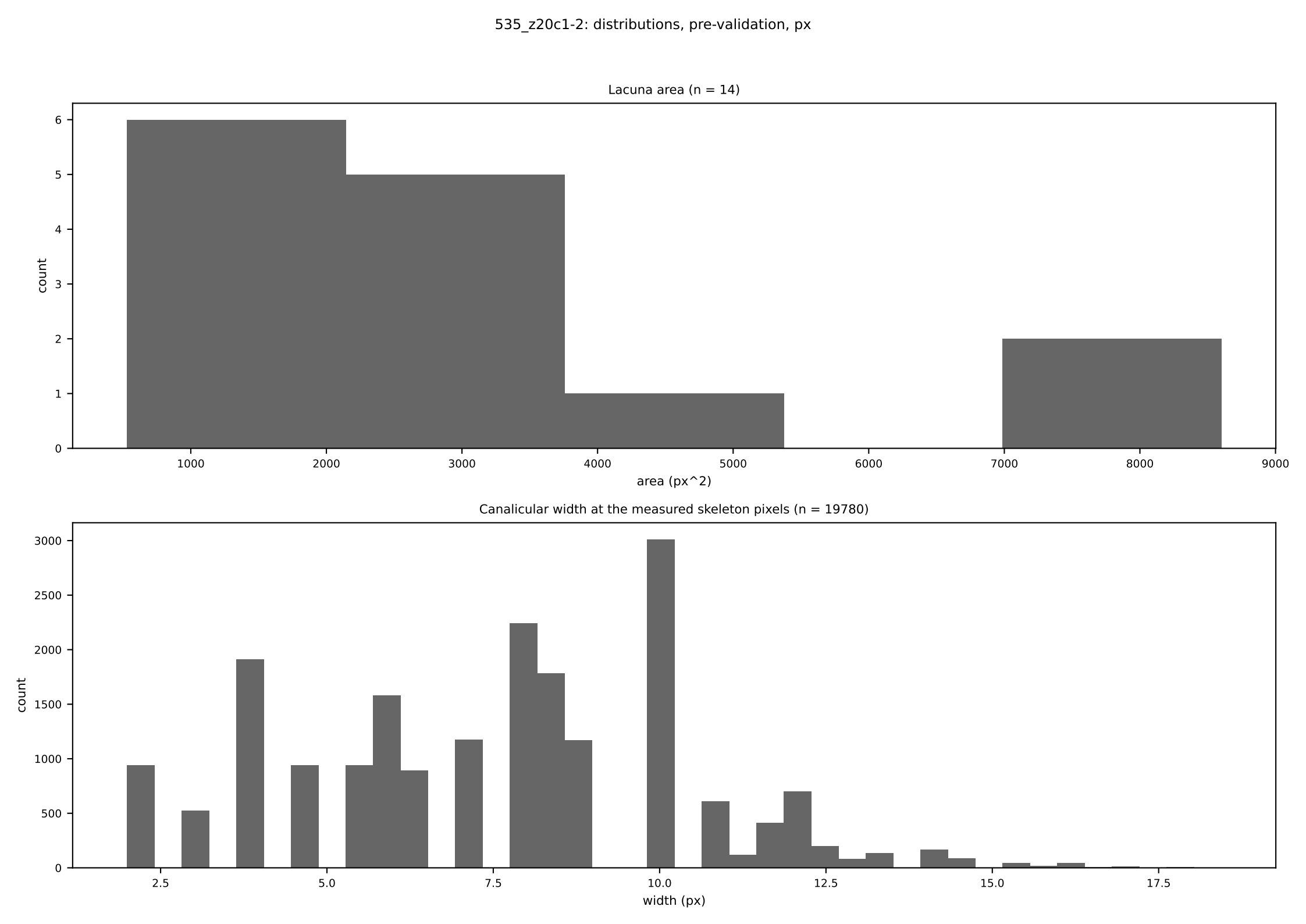
4	7	259	488	676.5	31	21.8226	9.2021
5	16	427	813	1139.9	62	18.3855	8.3815
6	11	290	651	872.382	34	25.6583	8.3643
7	11	391	733	1015.9	46	22.0849	7.96
8	7	227	616	354.865	19	18.6771	6.293
9	10	298	692	2233.25	78	28.6314	9.4006
10	7	229	366	330.735	17	19.455	8.07
11	2	152	334	746.441	16	46.6525	7.6066
12	18	425	849	773.215	37	20.8977	6.3092
13	8	349	763	1057.12	59	17.9173	7.3678
14	7	216	440	885.34	37	23.9281	7.5155

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	303.605	10283	25801	1	0.9954	0.034134	0.0323243	2.3056
2	92.8112	5363	16377	1	1	0.037852	0.0332173	5.3873
3	243.179	9488	24489	1	1	0.0419477	0.0378946	3.701
4	208.894	7431	16405	1	1	0.034854	0.029747	3.351
5	446.174	13512	24714	1	1	0.0316015	0.0328963	3.586
6	391.931	12275	26488	1	1	0.0236253	0.0245772	2.8066
7	312.191	11238	24088	1	1	0.0347927	0.0304301	3.5235
8	198.995	8411	21092	1	1	0.0269885	0.0292054	3.5177
9	327.019	11681	28795	1	1	0.0255115	0.0240319	3.0579
10	223.965	8975	20144	1	1	0.0255153	0.0181692	3.1255
11	139.681	5912	13025	1	1	0.0257104	0.025643	1.4318
12	832.034	18938	41375	1	1	0.0224417	0.0205196	2.1634
13	262.149	9124	20333	1	1	0.0382508	0.0375252	3.0517
14	177.782	8012	21351	1	1	0.0269596	0.0206079	3.9374

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	330	651	400.676	954.992	8	14	8
2	184	476	239.279	627.985	4	9	8
3	375	815	452.433	1047.05	11	10	11
4	259	449	298.593	563.63	6	7	9
5	353	563	478.99	918.394	8	16	14
6	250	480	320.593	731.612	12	9	7
7	332	663	445.99	846.666	8	13	12
8	125	217	255.693	689.257	6	7	10
9	220	465	343.948	781.583	6	11	10
10	226	356	256.551	415.505	6	8	6
11	90	198	172.782	372.421	2	7	5
12	421	571	471.12	963.122	12	6	6
13	349	550	402.291	872.352	5	10	9
14	211	328	237.782	490.676	5	5	5


535_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 14)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 19780)
count
width (px)